

PROJECT DEFENSE PREPARATION

Deliva — Likely Panel Questions & Model Answers

Organised by theme, hardest-to-remember material first within each section. Companion to the Technical Documentation.

How to use this: each question is collapsed by default — try answering it in your own words before clicking to reveal the model answer. That mirrors the actual defense better than reading answers straight through.

SECTIONS

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A. Motivation & scope 6 QUESTIONS

Q: What problem does this project solve, and for whom?

Q: Why model it on Bolt/Uber specifically?

Q: Why a delivery/logistics app rather than another domain?

Q: Who are the three user roles, and what can each one do?

Q: What was the single hardest technical problem in this project, and how did you solve it?

Q: What would you do differently if you started again?

B. System architecture 7 QUESTIONS

Q: Describe the overall architecture in one sentence.

Q: Is this a monolith, or microservices?

Q: What is server-side rendering (SSR), and does this app use it?

Q: Why Vite instead of, say, webpack or Create React App?

Q: What does "file-based routing" mean, concretely, in this project?

Q: What is Hono, and why is it used here instead of Express?

Q: Trace what happens, end to end, from clicking "Confirm Booking" to seeing the tracking ID.

C. Frontend / React 8 QUESTIONS

Q: What is a "controlled component," and where is it used here?

Q: What is a React Hook? Name three used in this project and what each does.

Q: Why does the autocomplete input use `useRef` for the debounce timer instead of `useState` ?

Q: What is "debouncing," in plain terms, and why does the address field need it?

Q: What does `AbortController` do, and why is it needed alongside debouncing?

Q: What's the difference between `onChange` and `onSelect` on the `PlaceAutocompleteInput` component?

Q: What is JSX, and is it valid JavaScript?

Q: Why is `key` used when rendering the list of suggestions?

D. Backend, API & database 7 QUESTIONS

Q: What is a REST API, and how does this project's `/api/deliveries` route fit that definition?

Q: Why is there no ORM (like Prisma or Sequelize) — just raw SQL?

Q: What is SQL injection, and how is this project protected against it?

Q: Why Postgres, and why specifically Neon's "serverless" flavor of it?

Q: What tables does this app's data model roughly consist of?

Q: How is a unique tracking ID generated for each delivery?

Q: What HTTP status codes does the API use, and why do they matter?

E. Authentication & security 6 QUESTIONS

Q: How does sign-in actually work in this app?

Q: What is `argon2`, and why hash passwords instead of storing them directly?

Q: What's the difference between hashing and encryption?

Q: What is a session, and where is it stored here?

Q: Is exposing an API key like `NEXT_PUBLIC_LOCATIONIQ_API_KEY` in the browser a security hole?

Q: What would you add to harden this app's security before a real production launch?

F. The map/autocomplete feature 8 QUESTIONS

Q: Why did the original Google Places plan fail, and what does that teach about dependency choices?

Q: What alternative was chosen, and why that one specifically?

Q: Is the address search "real-time"? What does that actually mean here?

Q: What data does selecting a suggestion actually capture, and where does it go?

Q: Why clear the stored lat/lng back to `null` every time the user types after selecting a suggestion?

Q: What would happen if a user typed an address and never selected a suggestion?

Q: How would you extend this to show pickup and delivery points on an actual map (not just a text dropdown)?

Q: What's the difference between geocoding and autocomplete, technically?

G. Validation & data integrity 5 QUESTIONS

Q: Walk through the Nigerian phone number regex, piece by piece.

Q: Why validate on the client at all, if it can be bypassed?

Q: Is this app's server-side validation of phone/name currently sufficient?

Q: Why require a first *and* last name rather than just "non-empty"?

Q: What's a case this validation might incorrectly reject, and is that acceptable?

H. General CS/web fundamentals 10 QUESTIONS

Q: What is an API?

Q: What is JSON, and why is it used for the request/response bodies here?

Q: What is the difference between `GET` and `POST` ?

Q: What does `async / await` do, and why is it used throughout this codebase?

Q: What is a Promise?

Q: What is a package manager, and what does `node_modules` contain?

Q: What is a "lock file" (e.g. `bun.lock`), and why does it matter?

Q: What is CSS, and what is Tailwind CSS specifically?

Q: What does "responsive design" mean, and where does this project apply it?

Q: What is version control, and what would `git` add to this project if it were initialized?

I. Limitations & future work 5 QUESTIONS

Q: What are the known limitations of this project, stated plainly?

Q: If you had one more week, what's the single highest-value improvement?

Q: How would you add automated tests to this project?

Q: How would this scale if usage grew significantly?

Q: Is this project production-ready as-is?

End of defense Q&A · see the companion Technical Documentation for the full code walkthrough